

Code-Layout, Readability and Reusability

Date	29 September 2026
Team ID	SWTID-2026-7674
Project Name	AI FAQ ASSISTANT API
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout	YES	Used Prettier with 4 spaces indentation in React.js & Python code
2	Proper File Structure	Files and folders are logically organize	YES	/frontend, /backend, /api, /database, /nlp_models - clear MVC structure followed
3	Meaningful Variable Names	Variables reflect their purpose clearly	YES	Example: userQuery, faqConfidenceScore, tamilTranslation, whatsappResponse - easy to understand
4	Function / Method Names	Functions are descriptively named	YES	getFAQAnswer(), translateToTamil(), checkConfidence(), sendWhatsAppReply()
5	Code Comments	Inline and block comments explain log	YES	Added comments for NLP logic, API integration, and confidence scoring logic for future developers
6	Modular Design	Code is split into reusable functions/m	YES	Separate modules: auth.js, nlpEngine.py, faqService.js, whatsappService.js - reusable
7	No Redundant Code	Duplicate or unused code is removed	YES	Removed duplicate FAQ fetching logic, used common apiCall() function, no dead code
8	Error Handling	Exceptions and errors are handled gra	YES	Try-catch for API failures, Fallback "Please contact admin" if confidence <70%, logs saved in MongoDB

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Authenticion module	Node.js,JWT,OTP	Used in Admin Dashboard login, Student login via WhatsApp & Web App. Same JWT validation for all APIs.	High
2	NLP & Query Processing Module	Python,OpenAI,GPT-3.5 API,spaCY	Reused for both Tamil and English queries, text & voice input, website & WhatsApp bot. One function processQuery() for all channels.	High
3	FAQ Search & Matching Module	Node.js,MySQL,pinocode vector DB	Reused for semantic search, keyword search, and admin search in dashboard. Same getFAQAnswer() logic everywhere	High
4	Translation Module	Python,Google,Translate API	Reused to translate Query (Tamil -> English for processing) and Answer (English -> Tamil for user). Used in both directions.	Medium
5	Notification & Whatsapp module	Node.js,Whatsapp Business Api	Reused for sending auto-replies, fallback messages, and admin alerts for unanswered queries. Same sendReply() function.	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Follows MVC architecture. Clean folder structure - /frontend (React), /backend (Node), /nlp (Python), /database. Prettier & ESLint used for consistent formatting.
Readability	5	Meaningful variable names like userQuery, faqConfidenceScore. Proper indentation (4 spaces). Inline comments for NLP logic, API calls, and confidence scoring. Easy for new developer to understand.
Reusability	5	High reusability - Auth module reused in Web + WhatsApp + Admin. NLP module processQuery() reused for Tamil/English & text/voice. DRY principle followed, no duplicate code.
Documentation / Comments	4	All major functions documented with comments. README file with setup steps, API documentation for /api/query & /api/faq. Code comments explain logic for Tamil translation & fallback mechanism
Overall Score	4.5	Code is production-ready, maintainable, modular, and scalable. Meets all quality standards - layout, readability, reusability, error handling. Can be extended to other college departments easily.